

A karmic map of the Earth

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What this is

A map of the world in which every point carries the accumulated weight of the cruelty that people have inflicted on other people at that spot, over the whole of recorded history, displayed as color at roughly one-kilometer resolution. The unit of account is a single human being. An event is not the fundamental record; a soul is. A massacre of four hundred is four hundred records that happen to share a place, a date, and a perpetrator. Where a name survives, the record carries the name. Where it does not, the record is a placeholder soul whose existence is inferred from a count.

Three decisions from the scoping conversation shape everything below. The score attaches to the ground where the act happened, not to the capital that ordered it or the homeland the victims came from. The score is not a death count but a weighted quantity in which the manner of suffering matters. And the categories in scope are the mass killing of noncombatants, structural cruelty such as slavery and engineered famine, and the punishment and crime tier: judicial torture and execution, lynching, and individual atrocity. Battlefield deaths between combatants are excluded.

"Human" means the genus *Homo*, not the species. A Neanderthal butchered at Moula-Guercy, the children of *Homo antecessor* consumed at Gran Dolina, and the individual struck twice on the forehead at Sima de los Huesos some 430,000 years ago are souls on this map. That decision, taken in review, pushes the start of the time axis back from the beginning of writing to nearly a million years before present, and it means the earliest entries rest entirely on bone rather than on any record.

The map should be understood as a memorial that happens to be quantitative, not a league table. That framing matters for what gets built: it argues for keeping names where they exist, for honesty about what is not recorded, and against any presentation that invites comparing one people's suffering to another's as a contest.

The soul as the unit of record

Every record describes one person and one episode of cruelty inflicted on that person. The same person can appear more than once if they suffered distinct episodes at distinct places, for example a deportation and then a camp. Each record has the following fields.

An identity block: a name if known, otherwise a generated placeholder identifier; sex and age if known; a flag for whether the identity is documented or inferred.

An episode block: the category (mass killing, structural, punishment or crime); the outcome (died, survived); the start and end dates of the suffering, with an uncertainty range; the perpetrator class (state, army, mob, institution, individual); and the source citation.

A place block: not a single coordinate but a footprint, which is a probability distribution over the surface of the Earth describing where this person's suffering occurred. For a named prisoner at a known camp the footprint is a point. For one of the two hundred thousand who died at Treblinka it is the same point. For an unnamed victim of the Bengal famine of 1943 it is a polygon covering the affected districts, weighted by

district mortality. For a person who died in the hold of a slave ship it is a line along the ship's route, weighted toward the later part of the voyage where mortality peaked.

A weight block: the cruelty multiplier, described in the next section, and the confidence in the record as a whole.

The reason to insist on this granularity even for events where nothing is known about individuals is that it keeps the arithmetic honest. Two events of ten thousand dead do not become one event of twenty thousand; they remain twenty thousand people, each of whom can be given a place and a weight, and the map is the sum over them. It also means that when a name list is later found for an event that was previously only a count, the placeholders are replaced one for one without changing the totals.

Scoring a single soul

The score of one record is a product of three factors. Written out:

$$s = O \times D \times M$$

Read aloud: the score s equals O times D times M . In words: the score for one person is the outcome factor, times the duration factor, times the manner factor.

The outcome factor O is 1 whether the person died or survived. The measure is accumulated pain over time, and death ends the accumulation rather than adding to it. A survivor may well go on suffering for decades, but the sources almost never record that, so the ledger scores only the documented episode and treats the two outcomes alike. The consequence is that the duration factor carries the entire distinction: a person shot within an hour of capture scores less than a person who survived a year in a ghetto, and the map will say so. The outcome is still stored on every record, so the ratio can be revisited without touching the data, but the default is parity.

The duration factor D captures how long the person suffered before the outcome. It is logarithmic in time, because the difference between an instant and a day matters more than the difference between year nine and year ten:

$$D = 1 + \log_{10}(1 + t / t_0)$$

Read aloud: D equals one plus the log base ten of one plus t over t -nought. In words: start at one, and add one for every factor of ten by which the suffering outlasted a reference time. The reference time t -nought is one day. A shooting has t near zero and D near 1. A year in a ghetto before deportation gives t of about 365 days and D of about 3.6. Thirty years of chattel slavery gives D of about 5. This is the factor that lets slavery and famine sit on the same axis as a massacre without needing a separate scoring system for structural cruelty.

The manner factor M captures intent and degradation. It is a small integer scale, set per event and inherited by every soul in the event unless individual records say otherwise. A value of 1 is negligent or indifferent killing, such as a famine allowed to run once it had begun. A value of 2 is deliberate killing carried out without additional degradation, such as execution by shooting. A value of 3 is deliberate killing or bondage with systematic degradation, such as the camp system, chattel slavery with its attendant violence, or torture preceding death. A value of 4 is reserved for cruelty in which the suffering was itself the purpose: torture for

its own sake, public torment as spectacle, sexual violence as a weapon. The scale is deliberately coarse. Finer gradations would imply a precision of moral judgment that the sources cannot support, and they would invite endless argument about individual cases.

A worked example. A person shot in a pit at Babyn Yar: O of 1, t of hours so D near 1, M of 2, score about 2. A person who spent eighteen months in the Warsaw ghetto and then died at Treblinka: two records. The ghetto record has D of about 3.7, M of 3, score about 11 at the ghetto's footprint. The Treblinka record has D near 1, M of 3, score 3 at the camp. The ghetto outweighs the gas chamber in this ledger, which is what a measure of accumulated suffering must say. A person enslaved in Virginia from age twelve to death at fifty: O of 1, D of about 5.1, M of 3, score about 15 spread over the plantation region where they were held. A person tortured to death over days in a medieval judicial proceeding: O of 1, D of about 1.5, M of 4, score 6.

The scores are small numbers by design. What makes a place dark on the map is not the weight of any one soul but the number of them. Auschwitz is 1.1 million records with score around 3 apiece, concentrated on a footprint of a few square kilometers. That is the point of soul-level resolution: the arithmetic is transparent and every term can be argued about separately.

Two things are deliberately not in the formula. There is no time decay; a cruelty in 500 BCE weighs the same as one in 1995. Karma in the sense the map intends does not fade, and a decay term would mostly encode the fact that old events are poorly recorded, which is better handled explicitly. And there is no offsetting credit for acts of mercy or rescue. A rescue layer could be built on the same schema and shown as a separate color channel, but subtracting it from the cruelty score would let a place launder its history, and the map should not permit that.

Placing a soul on the ground

The score of a map cell is the sum, over every soul, of that soul's score times the probability that their suffering occurred in that cell:

$$S(\text{cell}) = \sum \text{over souls of } s_i \times p_i(\text{cell})$$

Read aloud: S of cell equals the sum over i of s-sub-i times p-sub-i of cell. In words: each person contributes their whole score to the map, but spread across the cells of their footprint in proportion to how likely each cell is to be where it happened. The total over all cells of one person's contribution is exactly their score. Nothing is lost or double-counted by uncertainty about place; uncertainty only blurs.

Footprints come in four kinds. A point footprint is a single coordinate with a radius of a few hundred meters, used for camps, prisons, massacre sites, execution grounds, and lynching sites. A polygon footprint is a region with an optional density weighting, used for famines, pogrom waves, regional campaigns of extermination, and slaveholding regions; the density weighting comes from period population or from district-level mortality where sources give it. A route footprint is a line with a mortality profile along it, used for death marches, deportation rail lines, the Trail of Tears, and the Middle Passage. A city footprint is a polygon of the historical urban extent, used when a source says only that a massacre happened in a named city; a sack of a city in antiquity is the typical case.

For the slave trade specifically, one enslaved person generates up to four records at four footprints: capture in the interior, which is usually a broad polygon; confinement at the embarkation port, a point; the voyage,

a route; and the lifetime of bondage at the destination, a polygon over the plantation region. The voyage records carry the deaths at sea, which means the Atlantic itself will show color along the trade routes, as will the Mediterranean from the Barbary and Ottoman trades and the Indian Ocean from the Arab trade. Ocean cells are shown by default; the ocean is part of the Earth.

Placing a soul in time

Time has the same structure as place. A soul's episode occupies an interval, not an instant, and the interval is usually uncertain. The record therefore carries a temporal footprint: a probability distribution over dates describing when the suffering occurred, with the same rule that the whole score integrates to one across the footprint. A person shot at a known massacre has a footprint of one day. A person who died in the Great Leap famine has a footprint spread over 1959 to 1961 weighted by the provincial mortality curve. A person enslaved from 1820 to 1858 has a footprint that is uniform across those thirty-eight years, because the suffering was continuous, and their score is spread across the years in proportion. A victim of a massacre that a chronicle places only "in the reign of" a king has a footprint as wide as the reign.

The viewer's time control is then a window, defined by a start year and an end year, and the cell score shown is the sum over souls of their score times the fraction of their temporal footprint that falls inside the window. Written out:

$$S(\text{cell}, \text{window}) = \sum \text{over souls of } s_i \times p_i(\text{cell}) \times q_i(\text{window})$$

Read aloud: S of cell and window equals the sum over i of s-sub-i times p-sub-i of cell times q-sub-i of window. In words: each person contributes their score, spread over where it happened and over when it happened, and the viewer sees only the part that falls inside the chosen years. Set the window to all of history and the map is the full ledger. Narrow it to 1939 through 1945 and the enslaved person contributes nothing while the ghetto and the camp contribute everything. Narrow it to 1840 through 1850 and roughly a quarter of the enslaved person's score appears in Virginia while the camps vanish.

This is the primary defense against uneven recording. A viewer who suspects that Europe looks dark only because Europe kept records can drag the window to the fifteenth century and watch the modern documentation fall away, or step decade by decade and see where the record begins for each region. The documentation-density layer should respond to the same window so that the two can be read together.

The control itself should be a two-handled range slider on a nonlinear axis, because a linear axis from 3000 BCE to the present would put the entire twentieth century in the last two percent of the track. A log axis in years before present is the natural choice: it spreads antiquity, the medieval world, the early modern period, and the modern era into comparable stretches. Preset windows for the conventional periods, and a play control that sweeps a fixed-width window forward through time, are cheap and worth adding.

Computationally, the temporal footprint is discretized to bins whose width grows with the footprint's span: single years for anything dated to within fifty years, then five, fifty, five hundred and five thousand years. A camp dated to the month occupies one-year bins; a Neolithic massacre dated to a 150-year radiocarbon range occupies five-year bins; the Neanderthals at Krapina, dated only to a 10,000-year window, occupy five-hundred-year bins. The cell table is stored as cell by time-bin rather than cell alone, and the viewer computes the overlap of each bin with the chosen window. At resolution 8 with a few thousand bins this is a

large but sparse table, and most cells have entries in only a handful of bins. Pre-aggregating to coarser cells and coarser bins for the low-zoom, wide-window case keeps the interaction responsive.

The grid

The map is rendered at approximately one square kilometer per cell. Two options exist for the indexing scheme. A latitude-longitude grid of 0.01 degree is simple but its cells shrink toward the poles, so a Norwegian cell is a third the area of a Nigerian one and the color scale means different things in different places. The alternative, and the recommendation, is the H3 hexagonal hierarchy at resolution 8, whose cells average 0.74 square kilometers and vary in area by only a few percent across the globe. H3 also gives coarser parents for free, which is what a zoomable map needs: at continental zoom the display aggregates to resolution 4 or 5 and the sums roll up exactly.

The Earth's land surface is about 149 million square kilometers, so a dense resolution-8 grid over land is about 200 million cells. Nearly all of them will be zero, so storage is sparse: a table keyed by cell identifier holding the summed score, the soul count, and the record confidence. A hundred million soul records with footprints spread over polygons will generate contributions to perhaps ten million distinct cells, which is a modest table by modern standards. Placeholder souls in a large polygon do not need to be individually sampled; the polygon's total is distributed analytically by the density weighting, which is mathematically identical to sampling every soul and far cheaper.

The color scale must be logarithmic. Cell scores will span from single digits (one recorded murder) to several million (the extermination camps), and a linear scale would show the entire world as white with six black dots. Log base ten of cell score, mapped to a single-hue ramp from pale to deep, is the default. A perceptually uniform ramp is required so that a factor of ten looks like the same step everywhere on the scale. Whether to put a floor at one soul so that a single recorded murder is faintly visible, or to start the ramp at ten, is a design choice to make after seeing the prototype.

What the map cannot show, and how to be honest about it

The single largest distortion is that this is a map of recorded cruelty, and recording is uneven across time, place, and the literacy of the perpetrators and their victims. The camp system of the Third Reich is documented to the individual; the Mongol conquest of Khwarazm is documented by a handful of chroniclers who gave round numbers in the millions; the wars of pre-colonial sub-Saharan Africa and pre-Columbian Mesoamerica are documented mainly by the archaeology of mass graves and by the accounts of the colonizers who came after. A naive map will show Europe and China as the darkest places on Earth and much of Africa and the Americas as pale, and a viewer will draw the wrong conclusion.

Before the written record there is the skeletal one, and it deserves its own standing. Perimortem trauma, meaning injury at or around the time of death, has fracture edges and an absence of healing that distinguish it from damage after burial. Sharp-force, blunt-force and projectile wounds are separable, embedded points are common, and forearm fractures from warding off blows show a victim who saw it coming. What a single skeleton cannot show is the perpetrator or the motive. What an assemblage can show is a great deal: a pit of the old, the women and the children with blows to the back of the skull and no defensive wounds is an execution, not a battle; bound wrists, systematic mutilation or shins broken before death move the manner factor up the scale; healed wounds from earlier episodes on the same bodies, as at Jebel Sahaba, mean

recurrent raiding rather than one bad day. Forensic evidence is therefore neither recorded, since no one wrote it down, nor modeled, since it is not inferred from a population curve. It is a third confidence class, and the map carries it as such, with a contested flag for the sites where the violent reading is disputed, such as Krapina and Herxheim. Battlefield assemblages of combatants, Tollense, Towton, Visby and their kind, are excluded for the same reason combat deaths are excluded everywhere else.

The proposed remedy is three layers rather than one. The recorded layer is the default view and contains souls whose existence rests on a named list, a contemporaneous count, a modern scholarly estimate with stated methodology, or excavated remains with trauma. The modeled layer is an optional overlay, off by default, that adds souls whose existence is inferred, for example from demographic collapse where the mechanism is known to have included violence and enslavement; the Caribbean after 1492 is the canonical case. Each soul carries a confidence field, and the modeled layer's records carry low confidence and a stated model, and the overlay is drawn in a visibly different treatment, hatched or desaturated, so that it can never be mistaken for the record. The third layer is not a cruelty layer at all but a documentation-density layer: for each cell, how many independent sources, of what kind, underlie the score. Shown alongside the main map it tells the viewer where the map is trustworthy and where the pallor is ignorance rather than peace.

Estimated counts for events before about 1800 are contested, often by an order of magnitude. Each event record therefore carries a low, central, and high estimate for its soul count, and the pipeline can render any of the three. The default display uses the central estimate. The difference between the low and high renderings is itself informative and should be a toggle.

Sources, by tier

The first tier is name lists, where every record is a documented individual. Yad Vashem's central database holds roughly 4.9 million names of Shoah victims, many with place of death. The Arolsen Archives hold tens of millions of documents on persecution and forced labor under the Third Reich, keyed to individuals. Memorial's database of victims of Soviet political repression holds about three million names, and its companion gulag database maps several hundred camp administrations with sub-camps. The Documentation Center of Cambodia holds records on the Khmer Rouge period keyed to individuals and to the roughly two hundred prisons and twenty thousand mass graves it has mapped. The Srebrenica memorial, the Kigali genocide memorial, the Argentine and Chilean truth commissions, Guatemala's Recovery of Historical Memory project, and Peru's truth and reconciliation commission all published victim lists with places. In the United States, the Equal Justice Initiative and the earlier Tolnay-Beck inventory record about 4,400 racial terror lynchings by name, date, and county from 1877 to 1950. The Trans-Atlantic Slave Trade Database records about 36,000 voyages with embarkation and disembarkation ports, numbers embarked and disembarked, and therefore deaths at sea per voyage; its African Names Database holds about 91,000 individuals liberated from captured ships.

The second tier is georeferenced event datasets with counts but no names. The Uppsala Conflict Data Program's georeferenced event dataset covers organized violence from 1989 to the present with village-level coordinates and low, best, and high fatality estimates, and separately codes one-sided violence against civilians, which is exactly this map's mass-killing category. The Armed Conflict Location and Event Data project covers 1997 onward with similar precision. The Global Terrorism Database covers 1970 onward. The United States Holocaust Memorial Museum's Encyclopedia of Camps and Ghettos catalogs over 44,000 sites, each a point footprint. Andrew Walder's county-level dataset of deaths in the Chinese Cultural

Revolution, built from local gazetteers, and Cao Shuji's county-level estimates for the Great Leap famine give polygon footprints with counts for two of the largest events of the twentieth century.

The third tier is historical catalogs with regional footprints and contested counts. Peter Brecke's conflict catalog covers 1400 onward. Matthew White's compilation of atrocity death tolls, with its explicit attention to the range of estimates, is a useful index of pre-modern events and their sources even where its own numbers are not adopted. Wikidata holds several thousand items typed as massacre or genocide with coordinates and casualty fields, of very mixed quality; it is a discovery tool, not a source of record. For antiquity and the medieval world the sources are the chroniclers themselves, read through modern critical scholarship, and every such record will be modeled-layer with wide bounds.

Structural cruelty needs its own handling because the sources are censuses, not incident lists. The slaveholding population of the United States is known by county from the decennial censuses of 1790 to 1860, which gives a polygon footprint with density for roughly four million people at emancipation and perhaps ten million person-lifetimes over the institution's span. The Brazilian, Caribbean, and Spanish American slave populations are known less finely but adequately from colonial records. Serfdom in Russia, the Congo Free State's rubber regime, indentured labor in the Indian Ocean, the *encomienda* and *mita* systems, and the Ottoman and Arab slave trades all have scholarly population estimates that can be turned into placeholder souls with polygon footprints and duration factors. Engineered famines, meaning the Holodomor, the Great Leap, Bengal in 1943, the Irish famine to the extent policy prolonged it, and the Ethiopian famine of the 1980s, are scored with M of 1 or 2 depending on the historiography of intent, and their souls are placed by district mortality.

Frontier and settler-colonial violence has one exemplary dataset and otherwise almost none. Lyndall Ryan's *Colonial Frontier Massacres in Australia, 1788 to 1930*, records 438 sites of six or more killed, each with coordinates, a date, the recorded dead, victims, attackers, motive, weapons and a corroboration rating, and is the model for what such a source should look like. Nothing comparable exists for Aotearoa New Zealand, the Americas outside the lynching inventories, southern Africa or Siberia, so those regions enter through hand-curated tables and, for the demographic collapses that no one recorded body by body, through the modeled overlay.

The forensic tier has no database at all. It exists as site reports and a handful of syntheses: Meyer and colleagues on the Neolithic massacres of central Europe, Schulting and Fibiger's volume on skeletal trauma, Martin and Harrod on the bioarchaeology of violence, and the compilations behind Keeley's and Kissel and Kim's histories of early warfare. A curated table of a few hundred sites with individual counts, date ranges, trauma type and a contested flag covers essentially everything published with a body count, and the first fifty are in the prototype.

The individual-crime tier is the thinnest and the most biased. Judicial execution records exist for early modern England, France, and the German lands in some detail, and for imperial China through the dynastic histories; witch trials are catalogued for Europe with about fifty thousand executions and reasonably good place data. Serial murder and notable individual atrocity are recorded almost exclusively in literate modern societies with a press. This tier should be included because Kevin asked for it and because it is the only tier that makes the map speak at the resolution of a single life, but it will be visible only at high zoom and should be marked as such.

Pipeline

The build is a Python project run under uv, with the soul-record table stored as Parquet and queried with DuckDB. Each source gets an ingestion module that reads the source's native format, emits soul records in the common schema with footprints as geometries, and records provenance. Footprints are stored as WKT geometries plus a density raster reference where needed.

Scoring is a single pure function over a soul record and can be re-run in seconds when a parameter changes, which is the point of keeping the parameters separate from the data. Placement distributes each soul's score across H3 resolution-8 cells according to its footprint. For point footprints this is one cell or a small neighborhood. For polygons and routes it is a rasterization of the geometry onto the hex grid weighted by the density profile, done once per footprint and cached, since many thousands of souls share each footprint.

The output is a cell table at resolution 8 with parents pre-aggregated to resolutions 3 through 7. Rendering uses MapLibre with the cell table served as vector tiles, or deck.gl's H3 layer for a single-page prototype. The viewer gets the recorded layer by default with the modeled overlay and the documentation-density layer as switchable additions; a toggle for low, central, and high estimates; the two-handled time window described above; and a click action on any cell that lists the events and, where they exist, the names contributing to it. The last feature is what turns a heat map into a memorial and should be treated as a requirement rather than a nicety.

A static fallback is a single global GeoTIFF at resolution 8, rendered to PNG with the log ramp, for when the interactive map is not wanted.

Decisions taken and decisions still open

Settled in review: the outcome factor is 1 for death and survival alike; the manner scale has four steps; ocean cells are shown by default with a toggle to hide sea transits; the modeled layer is an optional overlay rather than part of the default view; the time control is a two-handled window with temporal footprints per soul; the genus Homo, not the species, defines a soul; and skeletal evidence forms a third confidence class, forensic, alongside recorded and modeled.

Still open: whether the documentation-density layer is a separate map or a transparency mask over the main map, which affects how the honesty caveat reads. Whether the time window defaults to all of history or opens on the modern era where the data is dense. Whether the manner factor for engineered famines should be fixed per event or vary by phase, since a famine that began through negligence and was then knowingly prolonged arguably moves from M of 1 to M of 2 partway through. And whether the first build includes the modeled overlay at all or leaves it as a documented gap for the second.

Phasing

The first build ingests five sources chosen to exercise every footprint type and both name-list and count-only records: the Trans-Atlantic Slave Trade Database for routes and ports, the Uppsala one-sided violence data for modern points with estimate ranges, the USHMM camp encyclopedia for dense historical points, the EJI lynching inventory for the individual tier, and Walder's Cultural Revolution counties for polygons with density. That set spans four centuries and four continents and will show whether the scoring and placement produce a map that reads correctly. Yad Vashem's names come second, since they are the largest

name list and turn the camp points from counts into people. Structural cruelty from census data and the pre-modern catalogs follow, and the modeled layer last.